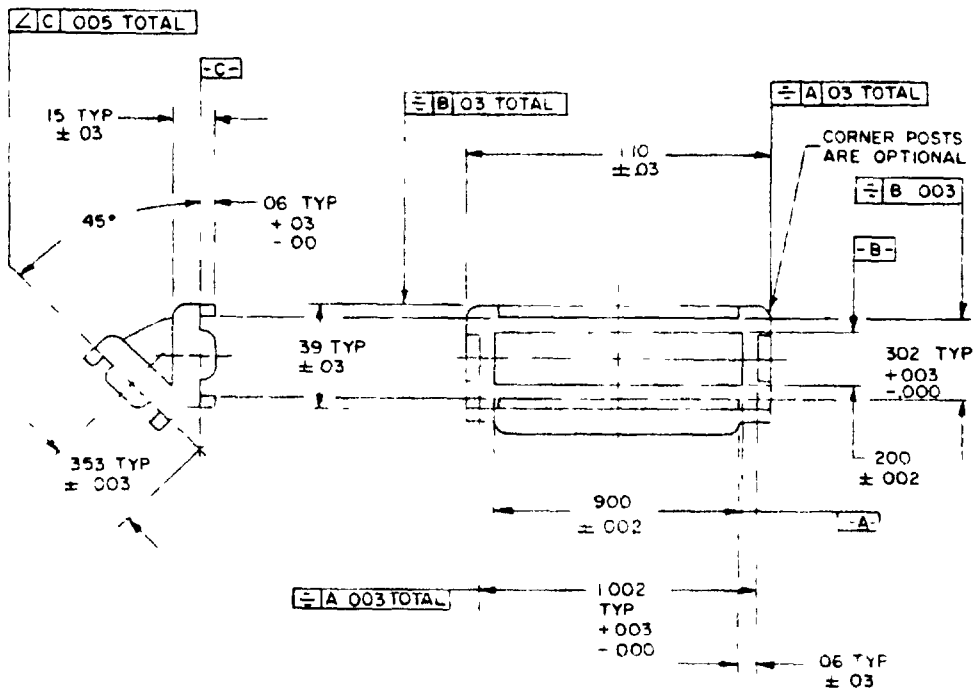
MILITARY SPECIFICATION SHEET

WAVEGUIDE ASSEMBLY, RIGID, CAST, 45-DEGREE E-PLANE BEND, REDUCED HEIGHT, FULL CORRAL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the waveguide assembly described herein shall consist of this document and the latest issue of Specification MIL-W-3970




| INCHES | MM | INCHES | MM | INCHES | MM |
|---|---|---|---|---|---|
| 002 | 05 | 06 | 1 5 | 353 | 8 97 |
| 003 | 08 | 15 | 3 8 | 39 | 9 9 |
| 005 | 13 | 200 | 5 08 | 900 | 22 86 |
| 03 | 8 | 302 | 7 67 | 1 002 | 25 45 |
| | | | | 1 10 | 27 9 |

NOTES
1 Dimensions are in inches
2 Metric equivalents are given for general information only and are based upon 1 00 inch = 25 4 mm

FIGURE 1 Dimensions for reduced height and full corral configuration

FSC 5985



Source: https://assist.dla.mil -- Downloaded: 2026-10-17T03:09Z
Check the source to verify that this is the current version before use.

REQUIREMENTS

Physical characteristics
Dimensions and configuration See figure ..
Operating temperature range -55° to +300°C
Material 6712 0, temper F, aluminum per QQ-A-[illegible] or 6712 1 per QQ-A-371
Surfaces (sandblasting after machining permissible)
Internal 63√ microinches per ANSI B46.1.

External 125√ microinches per ANSI B46.1.

Fungus resistance Materials nutrient to fungus shall not be used

Pressurization 45 lbf/in$^2$ gage maximum

Penetrant inspection No indication of any cracks at any location when tested per MIL-I-6866, type I, method P

Electrical characteristics
Frequency range 8.2 to 12.4 GHz.
VSWR 1.08 maximum
Power
Peak - 250 kW.
Average - 10 kW.

Part number M3970/24-001

NOTE The mating waveguides are 1/2 height of X-band waveguides, with a nominal wall thickness of .050 inch

Custodians
Army - EL
Navy - EC
Air Force - 85

Review activities
Army - MI,
Navy - EC, OS
Air Force - 11, 17, 99
DLA - ES

User activities
Army - AR
Navy - MC, CG, AS
Air Force - 14, 19

Preparing activity
Navy - EC

Agent
DLA - ES

(Project 5985-0881-4)



Source: https://assist.dla.mil -- Downloaded: 2026-10-17T03:09Z
Check the source to verify that this is the current version before use.

# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No 22 R255

INSTRUCTIONS The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay or will otherwise enhance use of the document DoD contractors, government activities or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government Fold on lines on reverse side staple in corner and send to preparing activity Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements Attach any pertinent data which may be of use in improving this document If there are additional papers, attach to form and place both in an envelope addressed to preparing activity

DOCUMENT IDENTIFIER AND TITLE
MIL-W- 3970/24

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3 IS THE DOCUMENT RESTRICTIVE?
☐ YES ☐ NO *(If "Yes" in what way?)*

4 REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

S/N 0102-014-1801

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T03:09Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE NAVY
NAVAL ELECTRONIC SYSTEMS COMMAND
WASHINGTON, D.C. 20360

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

POSTAGE AND FEES PAID
NAVY DEPARTMENT
DOD 316

COMMANDER
NAVAL ELECTRONIC SYSTEMS COMMAND
DEFENSE STANDARDIZATION PROGRAM BRANCH
DEPARTMENT OF THE NAVY
WASHINGTON, D.C. 20360

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T03:09Z
Check the source to verify that this is the current version before use.